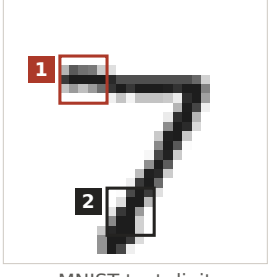
A Inspect the digit



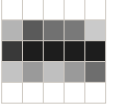
MNIST test digit

B Write the weights

−0.25	−0.25	−0.25	−0.25	−0.25
−0.25	−0.25	−0.25	−0.25	−0.25
+1	**+1**	**+1.5**	**+1**	**+1**
−0.25	−0.25	−0.25	−0.25	−0.25
−0.25	−0.25	−0.25	−0.25	−0.25

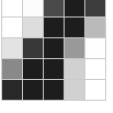
Kernel shown at 4× scale

C Apply the same detector



1 Horizontal stroke

ReLU(0.902) **= 0.902**



2 Diagonal stroke

ReLU(−0.098) **= 0**

ReLU(⟨W, X⟩ − 0.15) for patch X